

Notes: Hsieh and Klenow (QJE, 2009)

Misallocation and Manufacturing TFP in China and India

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1 Big picture

- **Question.** How much can cross-country productivity differences be explained by the misallocation of capital and labor across plants rather than by low technology inside a representative firm?
- **Setting.** The paper studies manufacturing plants in China, India, and the United States. It measures whether within narrowly defined industries some plants appear too small and others too large relative to their productivity.
- **Core challenge.** We never directly observe “distortions.” We only observe revenues, inputs, and industry output. The paper therefore needs a model that maps wedges in marginal products into observable revenue productivity.
- **Main theoretical result.** In a monopolistic-competition model with plant-specific output and capital distortions, efficient allocation implies equal TFPR within an industry. Dispersion in TFPR therefore measures misallocation.
- **Main quantitative result.** Moving to “U.S. efficiency” would raise manufacturing TFP by about 30% to 50% in China and 40% to 60% in India. Full equalization of TFPR within industries implies even larger gains: roughly 87% in China in 2005 and 128% in India in 1994.
- **Dynamic takeaway.** Allocative efficiency appears to improve in China between 1998 and 2005 but worsen in India between 1987 and 1994. The paper interprets this as evidence that Chinese reforms may have reduced misallocation, while Indian reforms had not yet visibly done so in the sample period.
- **Why this paper matters.** It turned the misallocation question into something measurable with plant-level data and became the benchmark paper for the “TFPR-dispersion” approach to misallocation.

2 Data and measurement (key objects)

2.1 Plant-level panels and country coverage

- **India.** The data come from the *Annual Survey of Industries* for fiscal years 1987–1988 through 1994–1995. The ASI is a census of larger registered manufacturing plants and a random one-third sample of smaller registered plants above the cutoff for registration.
- **China.** The data come from the *Annual Surveys of Industrial Production*, 1998–2005. The survey is a census of all state-owned firms plus all nonstate firms with revenue above 5 million yuan.
- **United States.** The benchmark data come from the *Census of Manufactures* for 1977, 1982, 1987, 1992, and 1997, dropping the smallest plants with limited production data.
- **Unit of observation.** A manufacturing plant or firm-year inside a four-digit manufacturing industry.
- **Observed variables.** Industry, labor compensation, value added, capital stock, age, and in some cases ownership and export status.

2.2 Constructed inputs and sample choices

- **Labor input.** The baseline labor measure is the wage bill, intended to proxy for effective labor by absorbing some variation in hours and worker quality.
- **Capital input.** Capital is measured from book values net of depreciation. In India and the United States, the paper averages beginning- and end-of-year book values when possible.
- **Industry shares.** Capital elasticities are pinned down using U.S. industry labor shares from the NBER Productivity Database, scaled up to account for nonwage compensation.
- **Trimming.** The paper trims the 1% tails of productivity and distortion measures within country-year cells to reduce the influence of outliers.
- **Scope.** The analysis is confined to manufacturing industries with reasonable counterparts across countries.

2.3 Physical versus revenue productivity

The paper distinguishes sharply between physical productivity and revenue productivity:

$$\text{TFPQ}_{si} \equiv A_{si} = \frac{Y_{si}}{K_{si}^{\alpha_s} (wL_{si})^{1-\alpha_s}}, \quad \text{TFPR}_{si} \equiv P_{si} A_{si} = \frac{P_{si} Y_{si}}{K_{si}^{\alpha_s} (wL_{si})^{1-\alpha_s}}.$$

- **TFPQ.** A physical-efficiency object: how many units of output a plant can produce from inputs.
- **TFPR.** A revenue-based object: physical productivity multiplied by the plant's price.
- **Key economic point.** In an undistorted industry, high-TFPQ plants should be larger and charge lower prices, so TFPR should be equalized across plants. Persistent dispersion in TFPR is therefore evidence of misallocation.

2.4 Main empirical objects

The empirical exercise measures:

- the dispersion of TFPQ within four-digit industries,
- the dispersion of TFPR within four-digit industries,
- the implied wedges in the marginal products of labor and capital,
- and the counterfactual TFP gains from reallocating labor and capital to reduce those wedges.

3 Models and empirical specifications (all equations + interpretation)

3.1 Final-good and industry aggregation

The final good is a Cobb–Douglas aggregate across manufacturing industries:

$$Y = \prod_{s=1}^S Y_s^{\theta_s}, \quad \sum_{s=1}^S \theta_s = 1.$$

Industry demand satisfies

$$P_s Y_s = \theta_s P Y.$$

Inside each industry, output is a CES aggregate of plant varieties:

$$Y_s = \left(\sum_{i=1}^{M_s} Y_{si}^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma-1}}.$$

- **Interpretation.** Plants are monopolistic competitors inside each industry, and σ is the elasticity of substitution across plant varieties.
- **Why it matters.** The CES structure implies a constant markup and a clean mapping from plant-level distortions to observable revenue productivity.

3.2 Plant production and distortions

Each plant has Cobb–Douglas technology:

$$Y_{si} = A_{si} K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}.$$

The model allows two plant-specific wedges:

- an **output distortion** $\tau_{Y,si}$, which scales both factor marginal products proportionally,
- and a **capital distortion** $\tau_{K,si}$, which changes the marginal product of capital relative to labor.

Profits are

$$\pi_{si} = (1 - \tau_{Y,si}) P_{si} Y_{si} - w L_{si} - (1 + \tau_{K,si}) R K_{si}.$$

- **Economic interpretation.** τ_Y captures things like output taxes, size restrictions, transport costs, or subsidies. τ_K captures wedges in the cost of capital such as privileged credit access or financing constraints.

3.3 Optimal input choice and marginal products

Profit maximization implies the pricing rule

$$P_{si} = \frac{\sigma}{\sigma-1} \left(\frac{R}{\alpha_s} \right)^{\alpha_s} \left(\frac{w}{1-\alpha_s} \right)^{1-\alpha_s} \frac{(1 + \tau_{K,si})^{\alpha_s}}{A_{si}(1 - \tau_{Y,si})}.$$

The capital-labor ratio satisfies

$$\frac{K_{si}}{L_{si}} = \frac{\alpha_s}{1 - \alpha_s} \cdot \frac{w}{R} \cdot \frac{1}{1 + \tau_{K,si}}.$$

Thus labor and output allocations are

$$L_{si} \propto \frac{A_{si}^{\sigma-1}(1 - \tau_{Y,si})^\sigma}{(1 + \tau_{K,si})^{\alpha_s(\sigma-1)}}, \quad Y_{si} \propto \frac{A_{si}^\sigma(1 - \tau_{Y,si})^\sigma}{(1 + \tau_{K,si})^{\alpha_s\sigma}}.$$

The marginal revenue products are

$$\begin{aligned} \text{MRPL}_{si} &= (1 - \alpha_s) \frac{\sigma - 1}{\sigma} \frac{P_{si} Y_{si}}{L_{si}} = w \frac{1}{1 - \tau_{Y,si}}, \\ \text{MRPK}_{si} &= \alpha_s \frac{\sigma - 1}{\sigma} \frac{P_{si} Y_{si}}{K_{si}} = R \frac{1 + \tau_{K,si}}{1 - \tau_{Y,si}}. \end{aligned}$$

- **Interpretation.** In an undistorted industry, the marginal revenue products of labor and capital should be equalized across plants. Distortions appear as cross-plant wedges in these objects.
- **Immediate implication.** Plants with subsidized capital or output support will have low observed marginal products; plants facing barriers will have high observed marginal products.

3.4 Why revenue productivity measures distortions

Since

$$\text{TFPR}_{si} \propto (\text{MRPK}_{si})^{\alpha_s} (\text{MRPL}_{si})^{1-\alpha_s} \propto \frac{(1 + \tau_{K,si})^{\alpha_s}}{1 - \tau_{Y,si}},$$

plant-level TFPR is a sufficient statistic for the combined wedge faced by the plant.

- **High TFPR.** The plant is too small relative to the efficient benchmark because it faces a barrier or wedge.
- **Low TFPR.** The plant is too large relative to its productivity because it is effectively subsidized.

Industry TFP is then

$$\text{TFP}_s = \left[\sum_{i=1}^{M_s} \left(A_{si} \cdot \frac{\text{TFPR}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{\frac{1}{\sigma-1}},$$

where TFPR_s is the geometric average of sectoral marginal revenue products.

If factor marginal products were fully equalized within the industry, efficient industry TFP would be

$$\bar{A}_s = \left(\sum_{i=1}^{M_s} A_{si}^{\sigma-1} \right)^{\frac{1}{\sigma-1}}.$$

Under a joint lognormality assumption,

$$\log \text{TFP}_s = \frac{1}{\sigma - 1} \log \left(\sum_{i=1}^{M_s} A_{si}^{\sigma-1} \right) - \frac{\sigma}{2} \text{Var}(\log \text{TFPR}_{si}).$$

- **Interpretation.** More dispersion in log TFPR means bigger misallocation and lower aggregate TFP.

3.5 Mapping the model to the data

The paper calibrates three key parameters as follows:

- $R = 0.10$, combining a 5% real interest rate and 5% depreciation.
- $\sigma = 3$, a conservative elasticity of substitution across plant varieties.
- α_s , set from U.S. industry labor shares, not from China or India, precisely because those countries may be distorted.

Plant-level wedges are backed out from observables as

$$1 + \tau_{K,si} = \frac{\alpha_s}{1 - \alpha_s} \frac{wL_{si}}{RK_{si}},$$

$$1 - \tau_{Y,si} = \frac{\sigma}{\sigma - 1} \frac{wL_{si}}{(1 - \alpha_s)P_{si}Y_{si}},$$

$$A_{si} = \kappa_s \frac{(P_{si}Y_{si})^{\sigma/(\sigma-1)}}{K_{si}^{\alpha_s} L_{si}^{1-\alpha_s}}.$$

- **Interpretation of τ_K .** High labor compensation relative to capital stock means capital is effectively expensive at that plant.
- **Interpretation of τ_Y .** Low labor share relative to the benchmark indicates an output wedge.
- **Interpretation of A_{si} .** Because only nominal output is observed, the paper infers real output from nominal revenue using the CES demand structure.

3.6 Counterfactual reallocation and measured gains

The aggregate ratio of actual to efficient output is

$$\frac{Y}{Y^{\text{efficient}}} = \prod_{s=1}^S \left[\sum_{i=1}^{M_s} \left(\frac{A_{si}}{\bar{A}_s} \frac{\text{TFPR}_s}{\text{TFPR}_{si}} \right)^{\sigma-1} \right]^{\theta_s/(\sigma-1)}.$$

- **Interpretation.** This is the key quantitative object. It says how much output is lost because plants do not all face the same effective marginal products.
- **Main benchmark.** The U.S. is treated as the comparison point because some dispersion in marginal products may reflect measurement error, adjustment costs, or omitted markup differences even in a relatively undistorted economy.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies misallocation in this paper

- **Model restriction.** The monopolistic-competition model implies that absent distortions, TFPR is equalized within a narrowly defined industry.

- **Observed dispersion.** Cross-plant dispersion in revenue productivity is therefore interpreted as dispersion in wedges, not just as dispersion in technology.
- **Narrow industries.** The paper works within four-digit sectors so that plants being compared are close substitutes technologically.
- **Benchmarking to the United States.** Since the model is not exact and the data are noisy, the authors do not ask for full equalization of marginal products. They ask how much China and India would gain if they only reduced dispersion to the level seen in the United States.

4.2 What limits identification

- **Measurement error.** Observed dispersion in TFPR could reflect noisy revenue or input measurement.
- **Adjustment costs.** Plants that are adjusting dynamically may temporarily have high or low marginal products even in an undistorted world.
- **Markup heterogeneity.** If bigger plants have higher markups, TFPR variation may partly reflect market power rather than policy wedges.
- **Capital-share heterogeneity.** Variation in capital-labor ratios inside an industry could reflect technology differences rather than distortions.
- **Fixed aggregate capital.** The baseline counterfactual keeps the aggregate capital stock fixed, so it is a partial-equilibrium efficiency exercise.

4.3 Why the U.S. benchmark is central

- The paper never claims that the United States is perfectly efficient.
- Instead, the United States is a pragmatic lower bound on how much residual dispersion may arise even without major distortions.
- Therefore the “move to U.S. efficiency” calculation is intentionally conservative relative to the full equalization benchmark.

4.4 Relation to the literature

- **Restuccia–Rogerson.** Provides the theoretical starting point that idiosyncratic wedges can depress aggregate productivity.
- **Chari–Kehoe–McGrattan.** Motivates the broader strategy of inferring distortions from residuals in first-order conditions.
- **Foster–Haltiwanger–Syverson.** Supplies the distinction between physical productivity and revenue productivity.
- **Banerjee–Duflo.** Motivates the focus on large marginal-product gaps in India.

5 Tables and figures

5.1 Figure I and Table I: dispersion of TFPQ

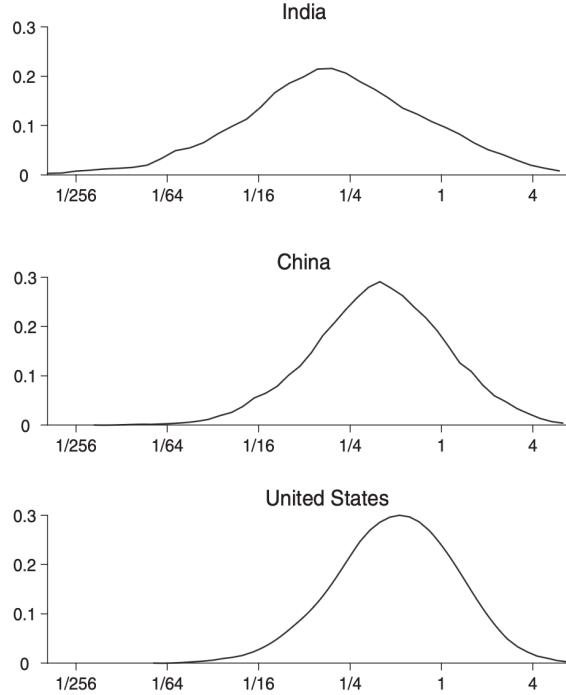


FIGURE I
Distribution of TFPQ

TABLE I
DISPERSION OF TFPQ

	1998	2001	2005
China			
S.D.	1.06	0.99	0.95
75 – 25	1.41	1.34	1.28
90 – 10	2.72	2.54	2.44
N	95,980	108,702	211,304
India	1987	1991	1994
S.D.	1.16	1.17	1.23
75 – 25	1.55	1.53	1.60
90 – 10	2.97	3.01	3.11
N	31,602	37,520	41,006
United States	1977	1987	1997
S.D.	0.85	0.79	0.84
75 – 25	1.22	1.09	1.17
90 – 10	2.22	2.05	2.18
N	164,971	173,651	194,669

Notes. For plant i in industry s , $TFPQ_{si} = \frac{Y_{si}}{K_{si}^{\alpha_K}(w_{si}L_{si})^{1-\alpha_L}}$. Statistics are for deviations of $\log(TFPQ)$ from industry means. S.D. = standard deviation, 75 – 25 is the difference between the 75th and 25th percentiles, and 90 – 10 the 90th vs. 10th percentiles. Industries are weighted by their value-added shares. N = the number of plants.

Read:

- Physical productivity dispersion is substantial in all three countries, but especially large in India.
- In the latest year, the ratio of the 75th to the 25th percentile of TFPQ is about 5.0 in India, 3.6 in China, and 3.2 in the United States.
- The left tail of the Indian TFPQ distribution is particularly thick, which the authors interpret as consistent with inefficient producers surviving longer than in the United States.
- This matters because the paper is not saying that China and India have no productive plants. It is saying that they also have too many plants with very low physical productivity still in operation.

5.2 Figure II and Table II: dispersion of TFPR

Read:

- TFPR dispersion is clearly larger in China and India than in the United States.
- In the latest year, the 75th/25th percentile ratio of TFPR is about 2.2 in India, 2.3 in China, and only 1.7 in the United States.
- The 90th/10th percentile ratio is about 5.0 in India, 4.9 in China, and 3.3 in the United States.

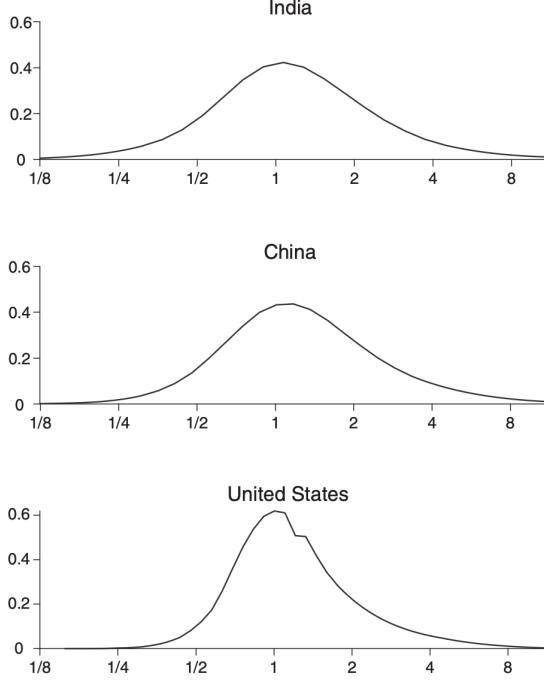


FIGURE II
Distribution of TFPR

TABLE II
DISPERSION OF TFPR

China	1998	2001	2005
S.D.	0.74	0.68	0.63
75 – 25	0.97	0.88	0.82
90 – 10	1.87	1.71	1.59
India	1987	1991	1994
S.D.	0.69	0.67	0.67
75 – 25	0.79	0.81	0.81
90 – 10	1.73	1.64	1.60
United States	1977	1987	1997
S.D.	0.45	0.41	0.49
75 – 25	0.46	0.41	0.53
90 – 10	1.04	1.01	1.19

Notes. For plant i in industry s , $TFPR_{si} = \frac{P_{si} Y_{si}}{K_{si}^{\alpha} (w_{si} L_{si})^{1-\alpha}}$. Statistics are for deviations of $\log(TFPR)$ from industry means. S.D. = standard deviation, 75 – 25 is the difference between the 75th and 25th percentiles, and 90 – 10 the 90th vs. 10th percentiles. Industries are weighted by their value-added shares. Number of plants is the same as in Table I.

- This is the central reduced-form evidence for misallocation: much larger cross-plant wedges in marginal products inside industries in China and India.

5.3 Table III: observable sources of TFPR variation

Read:

- Ownership, age, size, and region explain only a small share of within-industry TFPR dispersion.
- In India, all four sets of dummies together explain less than 5% of the variance of TFPR.
- In China, they explain only about 10%.
- So the bulk of dispersion is not captured by simple compositional differences; it is within ownership groups, age groups, and locations.

TABLE III
PERCENT SOURCES OF TFPR VARIATION WITHIN INDUSTRIES

	Ownership	Age	Size	Region
India	0.58	1.33	3.85	4.71
China	5.25	6.23	8.44	10.01

Notes. Entries are the cumulative percent of within-industry TFPR variance explained by dummies for ownership (state ownership categories), age (quartiles), size (quartiles), and region (provinces or states). The results are cumulative in that “age” includes dummies for both ownership and age, and so on.

TABLE IV
TFP GAINS FROM EQUALIZING TFPR WITHIN INDUSTRIES

China	1998	2001	2005
%	115.1	95.8	86.6
India	1987	1991	1994
%	100.4	102.1	127.5
United States	1977	1987	1997
%	36.1	30.7	42.9

Notes. Entries are $100(Y_{\text{efficient}}/Y - 1)$ where $Y/Y_{\text{efficient}} = \prod_{s=1}^S (\sum_{i=1}^{M_s} \frac{A_{si}}{A_s} \frac{TFPR_{si}}{TFPR_{\text{eff}}})^{y_s/(1-\alpha)}$ and $TFPR_{si} = \frac{P_{si} Y_{si}}{K_{si}^{\alpha} (w_{si} L_{si})^{1-\alpha}}$.

5.4 Table IV: gains from fully equalizing TFPR within industries

Read:

- Full within-industry equalization of TFPR would raise manufacturing TFP by 115.1%, 95.8%, and 86.6% in China in 1998, 2001, and 2005.
- The corresponding gains for India are 100.4%, 102.1%, and 127.5% in 1987, 1991, and 1994.
- Even the United States would gain 30% to 43% under full equalization, which is why the paper uses the United States as a benchmark rather than as a frictionless ideal.
- These are deliberately heroic upper bounds because they ignore measurement error and model misspecification.

5.5 Figure III and Table V: the efficient size distribution

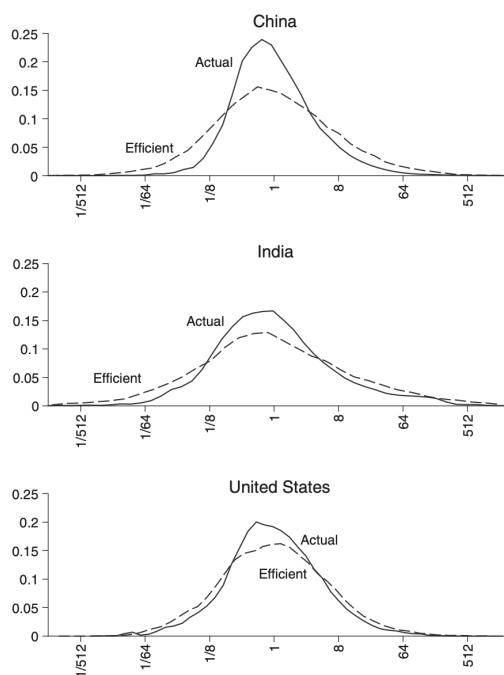


FIGURE III
Distribution of Plant Size

TABLE V
PERCENT OF PLANTS, ACTUAL SIZE VS. EFFICIENT SIZE

China 2005	0–50	50–100	100–200	200+
Top size quartile	7.0	6.1	5.4	6.6
2nd quartile	7.3	5.9	5.3	6.6
3rd quartile	8.5	6.0	5.2	5.4
Bottom quartile	10.5	5.9	4.5	4.2
India 1994	0–50	50–100	100–200	200+
Top size quartile	8.7	4.7	4.6	7.1
2nd quartile	10.7	4.6	4.1	5.7
3rd quartile	11.4	5.0	4.0	4.7
Bottom quartile	13.8	3.9	3.3	3.8
United States 1997	0–50	50–100	100–200	200+
Top size quartile	4.4	10.0	6.7	3.9
2nd quartile	4.4	9.6	5.8	5.1
3rd quartile	4.5	9.8	5.4	5.4
Bottom quartile	4.7	12.0	4.3	4.1

Notes. In each country-year, plants are put into quartiles based on their actual value-added, with an equal number of plants in each quartile. The hypothetically efficient level of each plant's output is then calculated, assuming distortions are removed so that TFPR levels are equalized within industries. The entries above show the percent of plants with efficient/actual output levels in the four bins 0%–50% (efficient output less than half actual output), 50%–100%, 100%–200%, and 200%+ (efficient output more than double actual output). The rows add up to 25%, and the rows and columns together to 100%.

Read:

- The efficient size distribution is more dispersed than the actual one in all three countries.
- In China and India, many plants of every initial size would shrink, especially in the 0%–50% bin where efficient output is less than half of actual output.
- But initially larger plants are more likely to expand than smaller plants in China and India, much more so than in the United States.
- This means large plants in China and India tend to have high marginal products and should be even larger, while many inefficient middle-sized and small plants survive at too large a scale.

TABLE VI
TFP GAINS FROM EQUALIZING TFPR RELATIVE TO 1997 U.S. GAINS

China	1998	2001	2005
%	50.5	37.0	30.5
India	1987	1991	1994
%	40.2	41.4	59.2

Notes. For each country-year, we calculated $Y_{\text{efficient}}/Y$ using $Y/Y_{\text{efficient}} = \prod_{s=1}^S [\sum_{i=1}^{M_s} (\frac{A_{si}}{A_s})^{\frac{\sigma}{\sigma-1}}]^{(\sigma-1)/\sigma}$ and $\text{TFPR}_{si} = \frac{P_{si} Y_{si}}{K_{si}^{\alpha} (w_{si} L_{si})^{1-\alpha}}$. We then took the ratio of $Y_{\text{efficient}}/Y$ to the U.S. ratio in 1997, subtracted 1, and multiplied by 100 to

5.6 Table VI: gains relative to U.S. efficiency

Read:

- Relative to the United States in 1997, moving to “U.S. efficiency” would raise Chinese manufacturing TFP by 50.5% in 1998, 37.0% in 2001, and 30.5% in 2005.
- For India the corresponding numbers are 40.2% in 1987, 41.4% in 1991, and 59.2% in 1994.
- The Chinese numbers imply that allocative efficiency improved by about 15% between 1998 and 2005, or about 2.0% per year.
- The Indian numbers imply deterioration of about 12% between 1987 and 1994, or about 1.8% per year.

5.7 Output growth and TFP-gap interpretation

Read:

- Comparing these changes to outside estimates of actual TFP growth, the paper suggests that better allocation may account for roughly one-third of China’s manufacturing TFP growth.
- The authors also estimate that misallocation might explain about 49% of the U.S.–China manufacturing TFP gap and about 35% of the U.S.–India gap.
- Because the average capital share is around one-half in China and India, long-run output gains with endogenous capital accumulation are much larger than the static TFP gains.
- A 30% TFP gain in China could imply a 67% long-run output gain; a 59% TFP gain in India could imply a 153% long-run output gain.

5.8 Robustness around the baseline gains

Read:

- Using age-adjusted current-market-value capital stocks leaves the main results essentially unchanged: 29.8% for China versus 30.5% baseline, and 59.9% for India versus 59.2% baseline.
- Using employment instead of the wage bill actually slightly lowers the gains: 25.6% in China and 57.4% in India relative to the U.S. benchmark.
- The results are highly sensitive to the within-industry elasticity σ . Raising σ from 3 to 5 boosts full-equalization gains from 86.6% to 184% in China and from 127.5% to 230% in India.

- The results are much less sensitive to the elasticity of substitution across sectors. Making sectors more complementary reduces gains modestly, while making them more substitutable raises gains modestly.

5.9 Tables VII–X: measurement-error checks

TABLE VII
TFP BY OWNERSHIP

	TFPR	TFPQ
China		
State	−0.415 (0.023)	−0.144 (0.090)
Collective	0.114 (0.010)	0.047 (0.013)
Foreign	−0.129 (0.024)	0.228 (0.040)
India		
State (central)	−0.285 (0.082)	0.717 (0.295)
State (local)	−0.081 (0.063)	0.425 (0.103)
Joint public/private	−0.162 (0.037)	0.671 (0.085)

Notes. The dependent variable is the deviation of log TFPR or log TFPQ from the industry mean. The independent variables for China are dummies for state-owned plants, collective-owned plants (plants jointly owned by local governments and private parties), and foreign-owned plants. The omitted group is domestic private plants. The independent variables for India are dummies for a plant owned by the central government, a plant owned by a local government, and a plant jointly owned by the government (either central or local) and by private individuals. The omitted group is a privately owned plant (both domestic and foreign). Regressions are weighted least squares with industry value-added shares as weights. Entries are the dummy coefficients, with standard errors in parentheses. Results are pooled for all years.

TABLE VIII
REGRESSIONS OF EXIT ON TFPR, TFPQ

China	
Exit on TFPR	−0.011 (0.003)
Exit on TFPQ	−0.050 (0.002)
India	
Exit on TFPR	−0.019 (0.005)
Exit on TFPQ	−0.027 (0.004)
United States	
Exit on TFPR	−0.011 (0.003)
Exit on TFPQ	−0.039 (0.002)

Notes. The dependent variables are dummies for exiting plants. The independent variables are the deviation of log(TFPR) or log(TFPQ) from the industry mean. Regressions are weighted least squares with the weights being industry value-added shares. Entries above are the coefficients on log(TFPR) or log(TFPQ), with standard errors shown in parentheses. Regressions also include a quartic function of plant age. Results are pooled for all years.

Read:

- Trimming 2% tails instead of 1% tails lowers the gains, but not by nearly enough to explain the baseline results.
- State-owned Chinese plants have about 41% lower TFPR than private domestic plants, while foreign-owned plants have higher TFPQ but lower TFPR. In India, plants with public involvement also have lower TFPR.
- Lower TFPR predicts higher plant exit in all three countries, and the relationship is at least as strong in China and India as in the United States. This is reassuring because it suggests TFPR captures true profitability, not just noise.
- Direct measurement-error checks using regressions of revenue on inputs and vice versa, and using growth-rate dispersion, give mixed results rather than a clean story that China and India simply have noisier data.

TABLE IX
REGRESSIONS OF INPUTS ON REVENUE, REVENUE ON INPUTS

	China	India	United States
Inputs on revenue	0.98	0.96	1.01
Revenue on inputs	0.82	0.90	0.82

Notes. Entries are the coefficients from regressions of $\log P_d Y_{it}$ on $\log K_{it}^{\alpha} (w L_{it})^{1-\alpha}$ (revenue on inputs) and $\log K_{it}^{\alpha} (w L_{it})^{1-\alpha}$ on $\log P_d Y_{it}$ (inputs on revenue). All variables are measured relative to the industry mean, with industries weighted by value-added shares. Results are pooled for all years.

TABLE X
DISPERSION OF INPUT AND REVENUE GROWTH

	Inputs	Revenue
China		
S.D.	0.45	1.00
75 – 25	0.34	0.93
India		
S.D.	0.28	0.70
75 – 25	0.24	0.60
United States		
S.D.	0.68	0.43
75 – 25	0.43	0.32

Notes. Entries are the standard deviation (S.D.) and interquartile range (75 – 25) of $d \log P_d Y_{it}$ and $d \log K_{it}^{\alpha} (w L_{it})^{1-\alpha}$. All variables are measured relative to the industry mean, and with industries weighted by their value-added shares. Entries are pooled for all years.

5.10 Tables XI–XIII and Figures IV–V: policy distortions

TABLE XI
OWNERSHIP OF INDIAN AND CHINESE PLANTS

	1998	2001	2005
China			
Private domestic	15.9	37.4	62.5
Private foreign	20.0	21.7	21.9
State	29.0	18.5	8.1
Collective	35.1	22.4	7.5
India	1987	1991	2004
Private	87.7	88.4	92.4
State (central gov.)	3.3	3.3	2.4
State (local gov.)	3.9	3.0	2.8
Joint public/private	5.1	5.4	2.4

Notes. Entries are the percent of plants in each ownership category in the sector, where each sector is weighted by the value-added share of the sector. Each column adds to 100.

TABLE XII
REGRESSIONS OF SECTOR TFPR DISPERSION ON STATE OWNERSHIP IN CHINA

	1998	2001	2005	1998–2005
State ownership share	0.766 (0.165)	0.659 (0.153)	0.025 (0.213)	0.300 (0.080)
Year F.E.	NO	NO	NO	YES
Sector F.E.	NO	NO	NO	YES
N	406	408	407	5,237

Note. Entries are coefficients from regressions of the variance of log TFPR in sector s on the variance in sector s of an indicator variable for a state-owned plant. All regressions are weighted by the value-added weights of the sector. Standard errors are clustered by sector in the last column.

TABLE XIII
REGRESSION OF SECTOR TFPR DISPERSION ON DELICENSING AND SIZE RESTRICTIONS IN INDIA

	(1)	(2)	(3)
Delicensed 1991	–0.298 (0.117)		–0.298 (0.117)
Delicensed 1991 × post-1991	0.032 (0.036)		–0.056 (0.040)
Size restriction		0.368 (0.173)	
Delicensed 1991 × post 1991 × size restriction			0.415 (0.120)

Notes. The dependent variable is the variance of log TFPR in sector s in year t . Entries are coefficients on the following independent variables: (1) delicensed 1991: indicator for whether industry was delicensed in 1991; (2) delicensed 1991 × post 1991: product of an indicator for an industry delicensed in 1991 and an indicator for observations after 1991; (3) size restriction: % of value-added of an industry subject to restrictions for small firms and; (4) delicensed 1991 × post 1991 × size restriction: product of size restriction, indicator variable for observations after 1991, and a dummy variable for industries delicensed after 1991. All regressions include indicator variables for year 1987 through 1994 and are weighted by the value-added share of the sector. Regressions (1) and (3) also include a dummy for industries delicensed in 1985. The control group consists of industries not delicensed in either 1985 or 1991. Standard errors are clustered by sector. Number of observations = 2,644.

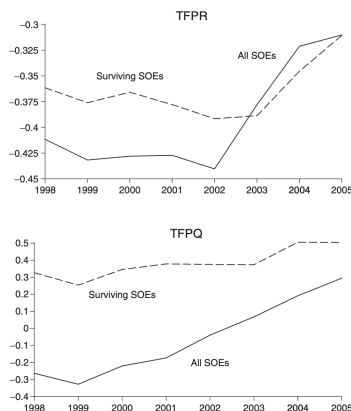


FIGURE IV
TFPR and TFPQ of Chinese State-Owned Firms

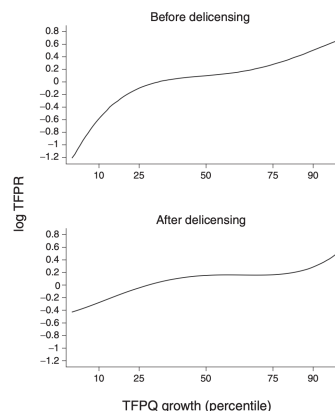


FIGURE V
TFPR and TFPQ Growth in Delicensed Sectors in India

Read:

- In China, state ownership shrinks sharply over time: about 29% of plants in 1998, 19% in 2001, and 8% in 2005.
- Across Chinese industries, a larger state-owned share is strongly associated with higher TFPR dispersion in 1998 and 2001, though not by 2005.
- Equalizing TFPR only within ownership groups explains a meaningful part of China's improvement in allocative efficiency: about 39% of the 1998–2005 decline in potential gains.
- In India, industries subject to size restrictions show higher TFPR dispersion, and after delicensing in 1991 the positive relation between TFPR and TFPQ growth flattens substantially, consistent with licensing having constrained efficient plants' expansion.

5.11 Figures VI–X and Table XIV: alternative explanations

Read:

- **Varying markups with size.** If larger plants everywhere had higher markups, one might expect TFPR to rise with size in all countries. Instead, TFPR rises strongly with size in India, mostly rises in China, and if anything falls with size in the United States.
- **Adjustment costs by age.** India shows TFPR increasing with age, not decreasing. China is mixed, while only the United States fits the idea that young plants have high TFPR because of adjustment costs.

- **Adjustment costs from input growth.** TFPR rises with input growth in all three countries, but the relationship is strongest in the United States, not in China or India.
- **Unobserved investments.** In the United States, low TFPR plants subsequently show higher TFPQ growth, consistent with unobserved investment. In China and India the pattern goes the other way, weakening that explanation.
- **Plant age and size composition.** Chinese plants are younger than U.S. plants, but Indian plants are older; Chinese plants are also much larger than U.S. plants, whereas Indian plants are smaller. Controlling for size and age explains only a modest fraction of the overall results.
- **Plant-specific factor shares.** Even when all within-industry capital-labor-ratio variation is reinterpreted as technology rather than distortions, large gains remain: about 23%–45% for China and 32%–39% for India relative to the U.S. benchmark.

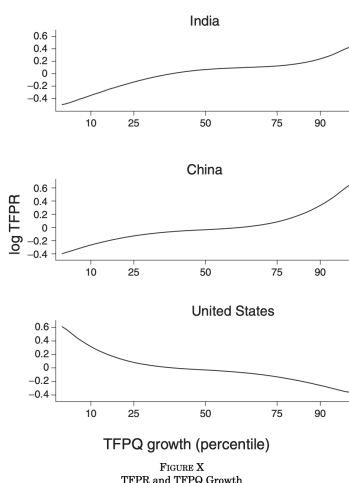
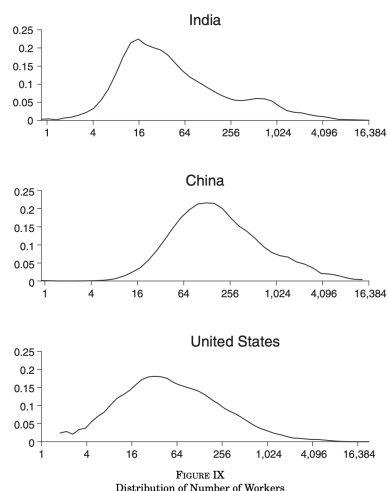
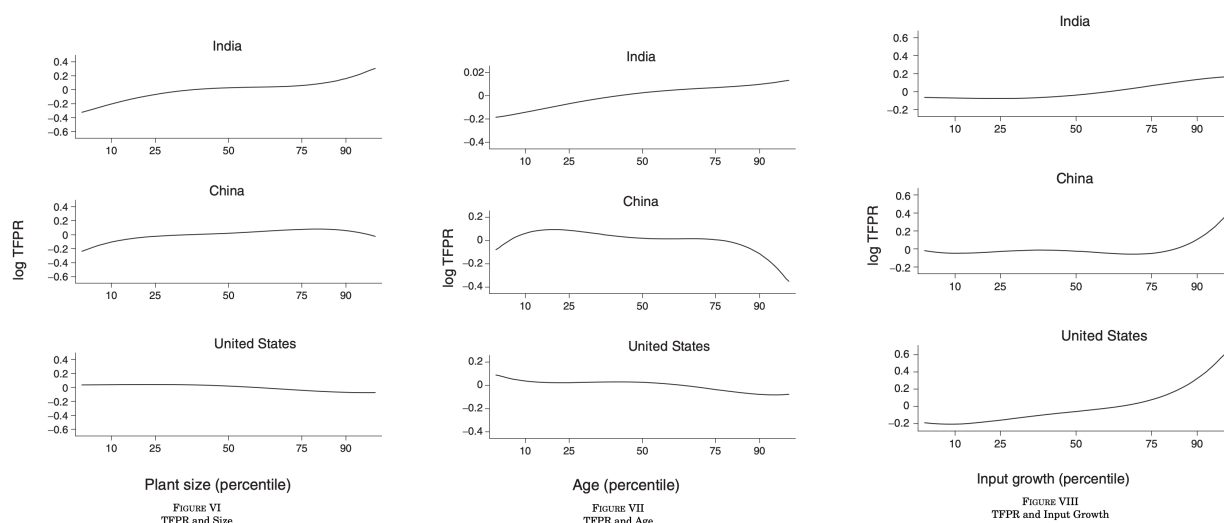


TABLE XIV
DISTRIBUTION OF PLANT AGE (PERCENTILES)

	25th	50th	75th
China	2	5	22
India	6	12	22
United States	5	10	25

Notes. Entries are the 25th, 50th, and 75th percentile distribution of plant age in a sector, where each sector is weighted by the value-added share of the sector.

6 Short summary

- **Conclusion.** Manufacturing TFP in China and India appears to be substantially depressed by within-industry misallocation of labor and capital across plants.
- **Identification insight.** The paper exploits the distinction between physical productivity and revenue productivity: in the model, dispersion in TFPR reveals wedges in marginal products.
- **Empirical lesson.** China improves over 1998–2005, India worsens over 1987–1994, and large measured gains remain even after numerous robustness and alternative-explanation exercises.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that one can use plant-level dispersion in revenue productivity to back out large, economically meaningful losses in manufacturing TFP from misallocation.
- The main quantitative headline is big: reducing within-industry wedges to U.S. levels would imply very large TFP gains in both China and India.
- The paper also shows that these losses are not just a static curiosity. Their time pattern lines up with broader reform narratives: improvement in China and worsening in India over the paper’s sample windows.

7.2 Economic intuition

- Efficient allocation means productive plants should be big and low-productivity plants should be small or exit.
- When policies or frictions subsidize some plants and penalize others, marginal products become unequalized. Some low-productivity plants employ too much capital and labor, while some high-productivity plants remain too small.
- The genius of the paper is to show that this misallocation leaves an observable footprint in TFPR: plants that are “too small” for their productivity look like high-TFPR plants, while those that are “too large” look like low-TFPR plants.

7.3 Takeaway

This paper is best read as the benchmark empirical misallocation paper. Its core message is that large productivity losses can arise not because every firm is individually inefficient, but because the economy places too many resources in the wrong firms. The paper’s framework is stylized and the authors are careful about caveats, but the central lesson is powerful and durable: large cross-plant wedges in marginal products are quantitatively important, measurable with microdata, and plausibly tied to policy distortions. For almost any later paper on misallocation, TFPR dispersion, or resource wedges, this is the starting point.